TWISTING A RELATIVELY AMPLE SHEAF BY A PULLBACK OF AN AMPLE SHEAF

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file morphisms.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- Separation and quasi-compactness consequences of ampleness and relative ampleness. Properties, Lemma \ref{properties-lemma-ample-separated} and Lemma \ref{lemma-relatively-ample-separated}, cited at proof lines 8845-8849. Give separatedness of the source and target.
- The criterion producing sections with affine nonvanishing locus. Properties, Lemma \ref{properties-lemma-invert-s-sections}, cited at proof lines 8858-8862. Produces the sections used at each point.

Source reference. Original lemma 0892: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10482–10482. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 01VH: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2446–2446. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 01PS: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2241–2241. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 09MP: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 12187–12187. Navigation pointer to a statement the

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counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 01VI: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2447–2447. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 01KU: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2063–2063. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 01PW: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2245–2245. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 01Q3: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2252–2252. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 0890: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10480–10480. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 0891: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10481–10481. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 01PT: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2242–2242. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let $f : X \to Y$ be a morphism of schemes, $\mathcal{M}$ an invertible $\mathcal{O}_Y$-module, and $\mathcal{L}$ an invertible $\mathcal{O}_X$-module.*

(1) *If $\mathcal{L}$ is f-ample and $\mathcal{M}$ is ample, then $\mathcal{L} \otimes f^*\mathcal{M}^{\otimes a}$ is ample for $a \gg 0$.*
(2) *If $\mathcal{M}$ is ample and f quasi-affine, then $f^*\mathcal{M}$ is ample.*

Proof. Assume $\mathcal{L}$ is f-ample and $\mathcal{M}$ ample. By assumption Y and f are quasi-compact (see Definition 01VH and Properties, Definition 01PS). Hence X is quasi-compact. By Properties, Lemma 09MP the scheme Y is separated and by Lemma 01VI the morphism f is separated. Hence X is separated by Schemes, Lemma 01KU. Pick $x \in X$. We can choose $m \geq 1$ and $t \in \Gamma(Y, \mathcal{M}^{\otimes m})$ such that Y_t is affine and $f(x) \in Y_t$. Since $\mathcal{L}$ restricts to an ample invertible sheaf on $f^{-1}(Y_t) = X_{f^*t}$ we can choose $n \geq 1$ and $s \in \Gamma(X_{f^*t}, \mathcal{L}^{\otimes n})$ with $x \in (X_{f^*t})_s$ with $(X_{f^*t})_s$ affine. By Properties, Lemma 01PW part (2) whose assumptions are satisfied by the above, there exists an integer $e \geq 1$ and a section $s' \in \Gamma(X, \mathcal{L}^{\otimes n} \otimes f^*\mathcal{M}^{\otimes em})$ which restricts to $s(f^*t)^e$ on X_{f^*t}. For any $b > 0$ consider the section $s'' = s'(f^*t)^b$ of $\mathcal{L}^{\otimes n} \otimes f^*\mathcal{M}^{\otimes (e+b)m}$. Then $X_{s''} = (X_{f^*t})_s$ is an affine open of X containing x. Picking b such that n divides $e + b$ we see $\mathcal{L}^{\otimes n} \otimes f^*\mathcal{M}^{\otimes (e+b)m}$ is the nth power of $\mathcal{L} \otimes f^*\mathcal{M}^{\otimes a}$ for some a and we can get any a divisible by m and big enough. Since X is quasi-compact a finite number of these affine opens cover X. We conclude that

for some a sufficiently divisible and large enough the invertible sheaf $\mathcal{L} \otimes f^*\mathcal{M}^{\otimes a}$ is ample on X. On the other hand, we know that $\mathcal{M}^{\otimes c}$ (and hence its pullback to X) is globally generated for all $c \gg 0$ by Properties, Proposition 01Q3. Thus $\mathcal{L} \otimes f^*\mathcal{M}^{\otimes a+c}$ is ample (Properties, Lemma 0890) for $c \gg 0$ and (1) is proved.

Part (2) follows from Lemma 0891, Properties, Lemma 01PT, and part (1). $\square$